October 19 Session
Concrete Examples of C^*-Algebras

1 Convention for the inner product

Let H be a complex Hilbert space and let (e_i) be an orthonormal basis of H. The convention used in the session is that the inner product is *linear in the first variable*. In particular,

$$\begin{aligned}\langle \alpha h_1 + \beta h_2, g\rangle &= \alpha\langle h_1, g\rangle + \beta\langle h_2, g\rangle,\\ \langle h, h\rangle &\geq 0, \qquad \langle h, h\rangle = 0 \iff h = 0,\\ \langle g, h\rangle &= \overline{\langle h, g\rangle}.\end{aligned}$$

Consequently, the second variable is conjugate-linear:

$$\langle h, \alpha g_1 + \beta g_2\rangle = \overline{\alpha}\,\langle h, g_1\rangle + \overline{\beta}\,\langle h, g_2\rangle.$$

2 Rank-one operators in $\mathcal{B}(H)$

Definition. For i, j, define the operator

$$e_i \otimes e_j : H \longrightarrow H, \qquad (e_i \otimes e_j)(h) := \langle h, e_j\rangle e_i.$$

In these notes the symbol $e_i \otimes e_j$ denotes this rank-one operator.

Its range is

$$(e_i \otimes e_j)(H) = \mathbb{C}e_i,$$

and therefore

$$\dim(e_i \otimes e_j)(H) = 1.$$

Thus, $e_i \otimes e_j$ is a rank-one operator.

Moreover, by the Cauchy–Schwarz inequality and because $\|e_i\| = \|e_j\| = 1$,

$$\begin{aligned}\|(e_i \otimes e_j)(h)\| &= \|\langle h, e_j\rangle e_i\|\\ &= |\langle h, e_j\rangle|\,\|e_i\|\\ &\leq \|h\|\,\|e_j\|\,\|e_i\|\\ &= \|h\|.\end{aligned}$$

In particular,

$$\|e_i \otimes e_j\| \leq 1, \qquad e_i \otimes e_j \in \mathcal{B}(H).$$

2.1 Adjoint of the operators $e_i \otimes e_j$

Write

$$T_{ij} := e_i \otimes e_j.$$

Then

$$T_{ij}^* = T_{ji}, \qquad \text{that is,} \qquad (e_i \otimes e_j)^* = e_j \otimes e_i.$$

Indeed, for $h, g \in H$,

$$\begin{aligned} \langle T_{ij} h, g \rangle &= \langle \langle h, e_j \rangle e_i, g \rangle \\ &= \langle h, e_j \rangle \langle e_i, g \rangle \\ &= \langle h, \langle g, e_i \rangle e_j \rangle \\ &= \langle h, T_{ji} g \rangle. \end{aligned}$$

By the definition of the adjoint, it follows that $T_{ij}^* = T_{ji}$.

2.2 Multiplication rule

For indices i, j, r, s,

$$(e_i \otimes e_j)(e_r \otimes e_s) = \delta_{jr}\, e_i \otimes e_s.$$

Indeed, for $h \in H$,

$$\begin{aligned} (e_i \otimes e_j)(e_r \otimes e_s)(h) &= (e_i \otimes e_j)\big(\langle h, e_s \rangle e_r\big) \\ &= \langle h, e_s \rangle \langle e_r, e_j \rangle e_i \\ &= \delta_{jr} \langle h, e_s \rangle e_i. \end{aligned}$$

This is exactly the multiplication rule satisfied by matrix units.

2.3 Composition with an arbitrary operator

Let $T \in \mathcal{B}(H)$. In the photographs, the following identities are left as exercises:

$$T \circ (e_i \otimes e_j) = T(e_i) \otimes e_j$$

and

$$(e_i \otimes e_j) \circ T = e_i \otimes T^*(e_j).$$

The second identity follows directly from

$$((e_i \otimes e_j)T)(h) = \langle Th, e_j \rangle e_i = \langle h, T^* e_j \rangle e_i.$$

3 Finite-rank projections

For each i,

$$p_i := e_i \otimes e_i$$

is a projection in $\mathcal{B}(H)$, since

$$p_i^* = p_i$$

and, using the multiplication rule,

$$p_i^2 = (e_i \otimes e_i)(e_i \otimes e_i) = e_i \otimes e_i = p_i.$$

Moreover,

$$p_i(H) = \mathbb{C} e_i,$$

so p_i has rank one.

3.1 The corner determined by a rank-one projection

Let $p_i = e_i \otimes e_i$ and let $T \in \mathcal{B}(H)$. Write

$$T(e_i) = \sum_r \lambda_r e_r.$$

Then the corner

$$p_i \mathcal{B}(H) p_i$$

consists of scalar multiples of p_i. In particular,

$$p_i \mathcal{B}(H) p_i = \mathbb{C} p_i \cong \mathbb{C}, \qquad \lambda p_i \longmapsto \lambda.$$

A direct way to identify the coefficient is

$$\begin{aligned} p_i T p_i(h) &= p_i T\big(\langle h, e_i\rangle e_i\big) \\ &= \langle h, e_i\rangle p_i T(e_i) \\ &= \langle h, e_i\rangle \left\langle \sum_r \lambda_r e_r, e_i \right\rangle e_i \\ &= \lambda_i \langle h, e_i\rangle e_i \\ &= \lambda_i p_i(h). \end{aligned}$$

Therefore,

$$p_i T p_i = \lambda_i p_i.$$

Exercise. If A is a C^*-algebra and $p \in A$ is a projection, verify that

$$pAp := \{pap : a \in A\}$$

is a C^*-subalgebra of A. In the photographs this subalgebra is called the *corner* determined by p.

In particular, when p is a rank-one projection in a representation on a Hilbert space,

$$pAp \cong \mathbb{C}$$

in the situation considered in class.

3.2 Rank of a projection

The terminology used in the session is

$$p \text{ has rank one} \quad \Longleftrightarrow \quad \dim p(H) = 1,$$

and, more generally,

$$p \text{ has rank } n \quad \Longleftrightarrow \quad \dim p(H) = n.$$

Example. Consider

$$p = e_1 \otimes e_1 + e_2 \otimes e_2.$$

Using the multiplication rule for matrix units,

$$\begin{aligned} p^2 &= (e_1 \otimes e_1 + e_2 \otimes e_2)^2 \\ &= e_1 \otimes e_1 + e_2 \otimes e_2 \\ &= p. \end{aligned}$$

Moreover, $p^* = p$, and

$$p(H) = \mathbb{C}e_1 + \mathbb{C}e_2 \cong \mathbb{C}^2.$$

Therefore, p is a rank-two projection.

4 The corner $p\mathcal{B}(H)p$ for a rank-two projection

Let

$$p = e_1 \otimes e_1 + e_2 \otimes e_2.$$

For $T \in \mathcal{B}(H)$, write the coordinates of T with respect to the orthonormal basis as

$$T(e_r) = \sum_k \lambda_{kr} e_k.$$

Then

$$\begin{aligned}
(e_i \otimes e_j)T(e_r \otimes e_s) &= (e_i \otimes e_j)\big(T(e_r) \otimes e_s\big) \\
&= (e_i \otimes e_j)\left(\sum_k \lambda_{kr} e_k \otimes e_s\right) \\
&= \sum_k \lambda_{kr}(e_i \otimes e_j)(e_k \otimes e_s) \\
&= \sum_k \lambda_{kr}\delta_{jk}\, e_i \otimes e_s \\
&= \lambda_{jr}\, e_i \otimes e_s.
\end{aligned}$$

Consequently,

$$\begin{aligned}
pTp &= (e_1 \otimes e_1 + e_2 \otimes e_2)T(e_1 \otimes e_1 + e_2 \otimes e_2) \\
&= (e_1 \otimes e_1)T(e_1 \otimes e_1) + (e_1 \otimes e_1)T(e_2 \otimes e_2) \\
&\quad + (e_2 \otimes e_2)T(e_1 \otimes e_1) + (e_2 \otimes e_2)T(e_2 \otimes e_2) \\
&= \lambda_{11}e_1 \otimes e_1 + \lambda_{12}e_1 \otimes e_2 + \lambda_{21}e_2 \otimes e_1 + \lambda_{22}e_2 \otimes e_2.
\end{aligned}$$

Thus every element of $p\mathcal{B}(H)p$ is determined by four complex coefficients. As a complex vector space,

$$\dim_{\mathbb{C}}(p\mathcal{B}(H)p) = 4.$$

A natural basis is

$$\{e_1 \otimes e_1,\ e_1 \otimes e_2,\ e_2 \otimes e_1,\ e_2 \otimes e_2\}.$$

The multiplication rules

$$(e_i \otimes e_j)(e_r \otimes e_s) = \delta_{jr} e_i \otimes e_s$$

are precisely the matrix-unit multiplication rules of $M_2(\mathbb{C})$.

Exercise. Let $S \in \mathcal{B}(H)$. Compute explicitly

$$pSp.$$

Then compute

$$(pSp)(pTp).$$

5 The commutative example $\mathbb{C}^n$

In the final part of the session, $\mathbb{C}^n$ is considered with coordinatewise operations. For

$$z = (z_i)_{i=1}^n, \qquad w = (w_i)_{i=1}^n,$$

define

$$zw = (z_i w_i)_{i=1}^n, \qquad z^* = (\overline{z_i})_{i=1}^n,$$

and the norm

$$\|z\|_\infty = \max_{1 \le i \le n} |z_i|.$$

With these operations, $\mathbb{C}^n$ is a unital commutative C^*-algebra with unit

$$1 = (1, \ldots, 1).$$

In the photographs this example is related to the representation theorem for commutative C^*-algebras:

$$\mathbb{C}^n \cong C(X),$$

and the following question is left open:

What is the space X in this example?